

Aleksandr Krasnok, Ph.D.

Assistant Professor, Electrical & Computer Engineering (FIU) | Director, Quantum Technologies & Metamaterials (QTM) Lab

IEEE Senior Member | Courtesy Appointment, Knight Foundation School of Computing and Information Sciences

Photonics • Quantum Optics • Quantum Technologies • Cryogenic Quantum Systems • Metamaterials

Florida International University
Miami, Florida

Email: akrasnok@fiu.edu

LinkedIn: [alexkrasnok](#)

Google Scholar: [profile](#)

Website: [QTM Lab](#)

ORCID: [0000-0001-7419-781X](#)

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Academic Profile

Assistant Professor of Electrical and Computer Engineering and Director of the Quantum Technologies & Metamaterials (QTM) Lab at Florida International University. Research in complex-frequency electrodynamics, nanophotonics, quantum control, sensing, and quantum-hardware diagnostics.

Research 200+ scholarly works; **15,222 citations**, *h-index* **61**, and *i*₁₀-**index** **113** (Google Scholar, September 25, 2026). Published in *Science*, *Nature*, *Physical Review Letters*, and journals in photonics and quantum technologies.

Quantum leadership

Co-founder, FIU Quantum Initiative; core member of the university-wide Committee on FIU Initiatives in Quantum Science, Education, and Industry Partnerships. Contribute quantum-hardware, photonics, and education expertise to planning for FIU's on-campus IonQ Superior 256 system, announced in September 2026 with arrival expected in late 2027.

Funding Established the FIU research program with a \$500,000 start-up package and separate support for three graduate students over two academic years. FIU subcontract PI for DOE/BNL El-Pho (\$600,000 ceiling; incremental funding); PI for AFOSR complex-frequency plasma research (\$50,000) and NSF I-Corps Meta-MRI (\$50,000). Additional research and computing support from ONR, the IC Postdoctoral Research Fellowship Program, RadiaBeam, and NVIDIA.

Education

Developed and taught FIU's undergraduate and graduate quantum-computing courses, EEE 4423 and EEE 6429, integrating Qiskit, Python, and open teaching materials. Author of *Introduction to Quantum Computing and Sensing for Electrical Engineers: A Hands-On Guide* (Springer, forthcoming 2026). Supervised three postdoctoral fellows; co-supervised one Ph.D. to completion and mentored undergraduate researchers and two Senior Design teams.

Service & honors

Associate Editor, *Photonics Research*; editorial boards of *Advanced Photonics Nexus* and *Reviews on Advanced Materials Science*. **2026 ASEMFL Rising Star in Engineering; Fulbright Specialist Roster, 2026–2029**; FIU Presidential Excellence Award finalist (2026); FIU Top Scholar (2025); Leopold B. Felsen Award (2024).

Professional Preparation

B.S., Quantum Optics , Far Eastern Federal University (FEFU), Vladivostok, Russia	2008
M.S., Quantum Optics , Far Eastern Federal University (FEFU), Vladivostok, Russia	2010
Ph.D., Photonics and Quantum Optics , ITMO University, Saint Petersburg, Russia	2013
Dissertation research in photonics, electrodynamics, and quantum optics.	
Postdoctoral Appointment, Photonics/Nanophotonics , The University of Texas at Austin, Austin, TX, USA	2016–2018

Appointments and Positions

Florida International University (FIU), Miami, FL, USA

Assistant Professor, Electrical & Computer Engineering

August 2021 – Present

Director, Quantum Technologies & Metamaterials (QTM) Lab; courtesy appointment, Knight Foundation School of Computing and Information Sciences (KFSCIS).

- **Research program and output.** Established the QTM Lab in complex-frequency electrodynamics, quantum photonics, cryogenic sensing, and quantum-hardware interfaces. Authored or coauthored **45+ journal publications since joining FIU in August 2021**, including work in *Science*, *Physical Review Letters*, and *Applied Physics Reviews*.
- **Funding and infrastructure.** Built the program with a **\$500,000 FIU start-up package**, plus separate tuition and stipend support for three graduate students over two academic years. FIU subcontract PI for DOE/BNL El-Pho (**\$600,000 ceiling; \$60,000 initially allotted**); PI for AFOSR (**\$50,000**) and NSF I-Corps (**\$50,000**); Co-PI/FIU lead for DOE SBIR/RadiaBeam (**\$63,000 to FIU**).
- **Collaborative experiments and theory.** Collaborations include Arjuna Madanayake's group at FIU and Andrea Alù's group at CUNY (USA); Sergey Kutsaev/RadiaBeam, Andrew Cleland/University of Chicago, and Michael Kelly/Argonne (USA); D. M. Broun/Simon Fraser University and Leonid Belostotski/University of Calgary (Canada); and the Sokoloff–Pascal–Callegari team at LAPLACE, Toulouse/CNRS (France). Joint work links circuit and electromagnetic modeling with superconducting-resonator and plasma experiments.
- **Education and quantum capacity.** Developed FIU's undergraduate/graduate quantum-computing sequence, EEE 4423 and EEE 6429, with Qiskit, Python, and open teaching resources. Co-founded the FIU Quantum Initiative and serve on its university-wide committee, contributing to research, education, and infrastructure planning for the announced IonQ campus installation.
- **Mentoring and career development.** Supervised three postdoctoral fellows and mentored 11 undergraduate researchers and Senior Design students. Co-supervised Deepanshu Trivedi (Ph.D., December 2025), now a Design Engineer with imec USA Florida's Superconducting Electronics team; former postdoc Muhammad Mahmudul Hasan joined Nokia in August 2026. Support technical independence, constructive feedback, and transitions to new academic and industry roles.

Institutional context. FIU rose from No. 78 to No. 46 among U.S. public universities in *U.S. News & World Report* (2022 edition; 2027 edition).

CUNY Advanced Science Research Center (ASRC), The Graduate Center, City University of New York, New York, NY, USA

Research Assistant Professor; Founding Director, Photonics Core Facility

July 2018 – July 2021

- **Facility creation and access.** Established the Photonics Core Facility's shared-user operations from the ground up, including experimental setups, operating procedures, user training, scheduling, and safety. Developed a facility with approximately **eight optical tables**, serving **100+ academic and industry users by 2021**.

- **Instrumentation.** Managed and developed capabilities around a Coherent Mira-HP Ti:sapphire femtosecond laser with an 18 W Verdi pump and second-/third-harmonic generation; Horiba iHR550 spectrometer; J.A. Woollam IR-VASE ellipsometer; NKT SuperK EXTREME supercontinuum source; visible/infrared cameras and spectrometers; and angle-resolved optical measurements.
- **Resources.** Managed a **\$162,200 operating and outfitting budget in FY2019**. Four major equipment items alone had documented historical acquisition costs exceeding **\$430,000**.
- **Research output and recognition.** Authored or coauthored **30+ journal publications during this appointment**, including *Nature*, *Nature Photonics*, *Nature Communications*, and *Physical Review Letters*. Contributed to the theory for the 2020 *Nature* study of photonic twistronics, selected among *Physics World's* Top 10 Breakthroughs of 2020.
- **International collaborations.** Worked with Andrea Alù/CUNY and Yuebing Zheng/UT Austin (USA), Cheng-Wei Qiu/National University of Singapore (Singapore), Qiaoliang Bao and Qingdong Ou/Monash University (Australia), and Jiaguang Han/Tianjin University (China). Connected modeling with optical and terahertz experiments, expanded shared experimental access, and contributed internationally recognized results to the ASRC's photonics program.

The University of Texas at Austin, Austin, TX, USA

Postdoctoral Researcher / Research Scientist

November 2016 – July 2018

- **Research output.** Authored or coauthored **20+ journal publications during this appointment** in Andrea Alù's group, spanning coherent absorption, nanoantennas, exciton-plasmon coupling, and quantum-dot control. Highlights include *Nature Reviews Materials* (2017), *Advanced Materials* (2018), and the Editors' Suggestion "Coherently Enhanced Wireless Power Transfer," *Physical Review Letters* (2018).
- **Theory-experiment collaboration.** Worked closely with Yuebing Zheng's experimental nanophotonics group at UT Austin, Yanwen Wu's group at the University of South Carolina, and Allan Bracker at the U.S. Naval Research Laboratory (USA). Joint work also involved Andrey Generalov/Aalto University (Finland), Denis Baranov/Chalmers University of Technology (Sweden), and collaborators at ITMO (Russia).
- **Contribution to the research program.** Developed analytical and numerical models for device design and interpretation of measurements, linking theoretical proposals to demonstrations in quantum-dot photonics, exciton-plasmon coupling, and coherent wireless power transfer. Helped extend the group's experimental collaborations and publication record.

ITMO University, Saint Petersburg, Russia

Senior Research Fellow, Photonics / Electrodynamics

June 2013 – November 2016

- **Group leadership.** Led an all-dielectric nanophotonics team of approximately **seven researchers: two postdoctoral fellows and five graduate/undergraduate students**. Directed projects on nonlinear, active, and quantum dielectric nanoantennas, including Russian Science Foundation-supported research; provided technical training and individual mentoring.
- **Research output.** Authored or coauthored **20+ journal publications during this appointment**, including *Nano Letters*, *Advanced Materials*, and *Scientific Reports*. Results advanced superdirective dielectric antennas, magnetic optical resonances, Purcell enhancement, and hybrid nanostructures.

- **International collaborations and experimental validation.** Worked with Pavel Belov's group and experimental researchers Sergey Makarov, Dmitry Zuev, and Ivan Mukhin in Russia; Yuri Kivshar and Andrey Miroshnichenko at the Australian National University (Australia); and Constantin Simovski and Sergei Tretyakov at Aalto University (Finland). Combined theory, simulations, fabrication, and optical/microwave measurements across the teams, developing ITMO's all-dielectric nanophotonics research and international partnerships.

Entrepreneurial Activities

- **Founder and Principal, SCIINSIGHT CONSULTING, LLC, Florida, USA** Feb 2026–Present

Research Interests

- **Complex-frequency electrodynamics and non-Hermitian scattering.** Pole-zero structure, complex-frequency driving, and waveform design for the control of resonant systems across RF, microwave, photonic, plasma, and quantum platforms.
- **Quantum control and quantum-hardware diagnostics.** Hardware-aware control and diagnostic methods for spectrally crowded quantum devices: complex-zero targeting, shaped-pulse spectroscopy, and physics-guided inference for superconducting and cryogenic resonant hardware.
- **Circuit and cavity QED, and photonic interfaces.** Multi-resonator architectures, Fano-type interference, impedance engineering, cryogenic packaging, scalable readout, and microwave-optical transduction.
- **Nanophotonics, metasurfaces, and reconfigurable antennas.** Dielectric nanoantennas, Mie-resonant metasurfaces, twist-tunable bilayers, time-modulated structures, and polaritonic systems for optical, RF, sensing, and communication applications.
- **Quantum sensing, cryogenic sensing, and applied metrology.** Resonant and quantum-enabled sensing across temperature stages: NV-diamond magnetometry, cryogenic dielectric resonators, RF and plasma diagnostics, MRI field-control metasurfaces, and infrastructure health monitoring.
- **Computational electromagnetics and quantum simulation.** Full-wave simulation, reduced-order and circuit models, QuTiP/Qiskit workflows, GPU and HPC computing, uncertainty quantification, and surrogate modeling linking theory to reproducible experiment.

Honors, Awards, and Recognitions

Major Honors and Recognitions

- **Rising Star in Engineering,** Academy of Science, Engineering and Medicine of Florida (ASEMFL), 2026. Selected September 2026; recognition at the November 2026 annual meeting.
- **Selected, Fulbright Specialist Roster,** U.S. Department of State, Bureau of Educational and Cultural Affairs. Roster term: August 14, 2026–August 14, 2029.

- **Finalist, FIU Presidential Excellence Award**, Florida International University, 2026. FIU recognition awards; FIU awards listing.
- **FIU Science & Fiction Lab Interdisciplinary Faculty Fellow**, Florida International University, 2026. Selected to develop an assessed interdisciplinary teaching module connecting engineering concepts with fiction-based analogies.
- **FIU Top Scholar Award (Research)**, Florida International University, 2024–2025. University-wide research recognition. CEC announcement; FIU listing.
- **Nominee, Blavatnik National Awards for Young Scientists**, Physical Sciences & Engineering, 2025. Institutional nomination. program.
- **Top 2% Scientist**, Stanford/Elsevier standardized citation-impact ranking, 2021–2025. doi:10.17632/btchxktzyw.8.
- **Best Scientists – Electronics & Electrical Engineering**, Research.com, 2025/2026. profile.
- **Leopold B. Felsen Award for Excellence in Electrodynamics**, EurAAP, 2024. award page.
- **IEEE Senior Member**, IEEE, 2024. definition.
- **APL Rising Stars**, *Applied Physics Letters*, 2023. Recognition for “Fano-qubits for quantum devices with enhanced isolation and bandwidth”, Appl. Phys. Lett. **122**, 264003. doi:10.1063/5.0151047.
- **Early-Career Award in Nanophotonics**, *Nanophotonics*, 2021. award page.
- **Top 10 Breakthrough of 2020**, *Physics World*. Recognition for photonic twistronics and twisted α -MoO₃ polaritons. Physics World; ASRC news.
- **Optics & Photonics News annual feature selection**, *Optics in 2018*. Recognition for “Coherent Optics for Energy Transfer, Storage and Release”. OPN link.
- **Editors’ Suggestion**, *Physical Review Letters*, 2018. Recognition for “Coherently Enhanced Wireless Power Transfer”, Phys. Rev. Lett. **120**, 143901. doi:10.1103/PhysRevLett.120.143901; Physics highlight.

Selected Earlier Distinctions

- **Gold Medal**, Nobel Laureate Zhores Alferov’s Foundation, 2016.
- **Best Young Scientist**, ITMO University, 2012.
- **Dynasty Foundation Scholarship**, multiple years.
- **St. Petersburg Government Grant for Young Scientists**, 2011.
- **Research Position Award**, Russian Federation Megagrant Program, 2011.

Selected Publications

Book

- **A. Krasnok**, *Introduction to Quantum Computing and Sensing for Electrical Engineers: A Hands-On Guide*. Synthesis Lectures on Emerging Engineering Technologies, Springer, Cham, first edition, forthcoming 2026.
Hardcover ISBN 978-3-032-34529-5; eBook ISBN 978-3-032-34530-1. [Springer book page](#).

Journal Articles

1. **A. Krasnok**, "Constant-Amplitude 2π Phase Modulation from Topological Pole–Zero Winding," *Physical Review Letters* **137**(5), 053801 (2026).
doi:10.1103/xt46-rfjs (single-author paper)
2. S. Kim, **A. Krasnok**, and A. Alù, "Complex-frequency excitations in photonics and wave physics," *Science* **387**(6741), eado4128 (2025).
doi:10.1126/science.ado4128 (field-setting review)
3. S. Kim, S. Lepeshov, **A. Krasnok**, and A. Alù, "Beyond bounds on light scattering with complex frequency excitations," *Physical Review Letters* **129**(20), 203601 (2022).
doi:10.1103/PhysRevLett.129.203601
4. **A. Krasnok**, D. G. Baranov, H. Li, M.-A. Miri, F. Monticone, and A. Alù, "Anomalies in light scattering," *Advances in Optics and Photonics* **11**(4), 892–951 (2019).
doi:10.1364/AOP.11.000892 (invited review)
5. **A. Krasnok**, D. G. Baranov, A. Generalov, S. Li, and A. Alù, "Coherently enhanced wireless power transfer," *Physical Review Letters* **120**(14), 143901 (2018).
doi:10.1103/PhysRevLett.120.143901 (Editors' Suggestion)
6. D. Trivedi, L. Belostotski, A. Madanayake, and **A. Krasnok**, "Fano-qubits for quantum devices with enhanced isolation and bandwidth," *Applied Physics Letters* **122**, 264003 (2023).
doi:10.1063/5.0151047 (APL Rising Stars Collection)
7. D. Trivedi, A. Madanayake, and **A. Krasnok**, "Revealing invisible scattering poles with complex-frequency signals," *Journal of Applied Physics* **137**, 243103 (2025).
doi:10.1063/5.0262971
8. **A. Krasnok**, P. Dhakal, A. Fedorov, P. Frigola, M. Kelly, and S. Kutsaev, "Superconducting microwave cavities and qubits for quantum information systems," *Applied Physics Reviews* **11**(1), 011302 (2024).
doi:10.1063/5.0155213
9. Q. Zhang, G. Hu, W. Ma, P. Li, **A. Krasnok**, R. Hillenbrand, A. Alù, and C.-W. Qiu, "Interface nano-optics with van der Waals polaritons," *Nature* **597**(7875), 187–195 (2021).
doi:10.1038/s41586-021-03581-5
10. G. Hu, Q. Ou, G. Si, Y. Wu, J. Wu, Z. Dai, **A. Krasnok**, Y. Mazor, Q. Zhang, Q. Bao, C.-W. Qiu, and A. Alù, "Topological polaritons and photonic magic angles in twisted α -MoO₃ bilayers," *Nature* **582**(7811), 209–213 (2020).
doi:10.1038/s41586-020-2359-9 (Physics World Top 10 Breakthroughs, 2020)

11. M. Kang, Z. Zhang, T. Wu, X. Zhang, Q. Xu, **A. Krasnok**, J. Han, and A. Alù, “Coherent full polarization control based on bound states in the continuum,” *Nature Communications* **13**, 4536 (2022). doi:10.1038/s41467-022-31726-1
12. S. Abdollahramezani *et al.* (incl. **A. Krasnok**), “Electrically driven reprogrammable phase-change metasurface reaching 80% efficiency,” *Nature Communications* **13**, 1696 (2022). doi:10.1038/s41467-022-29374-6
13. Z. L. Deng, T. Shi, **A. Krasnok**, X. Li, and A. Alù, “Observation of localized magnetic plasmon skyrmions,” *Nature Communications* **13**, 8 (2022). doi:10.1038/s41467-021-27710-w
14. M. Song, P. Jayathurathnage, E. Zanganeh, M. Krasikova, P. Smirnov, P. Belov, P. Kapitanova, C. Simovski, S. Tretyakov, and **A. Krasnok**, “Wireless power transfer based on novel physical concepts,” *Nature Electronics* **4**(10), 707–716 (2021). doi:10.1038/s41928-021-00658-x
15. D. G. Baranov, **A. Krasnok**, T. Shegai, A. Alù, and Y. Chong, “Coherent perfect absorbers: linear control of light with light,” *Nature Reviews Materials* **2**, 17064 (2017). doi:10.1038/natrevmats.2017.64

Invited perspectives / News & Views (selected):

- S. Lepeshov and **A. Krasnok**, “Tunable phase-change metasurfaces,” *Nature Nanotechnology* **16**(6), 615–616 (2021). doi:10.1038/s41565-021-00892-6
- **A. Krasnok**, “Metalenses go atomically thick and tunable,” *Nature Photonics* **14**(7), 409–410 (2020). doi:10.1038/s41566-020-0648-3

Strategic Initiatives and Engagements

- **Core Member, Committee on FIU Initiatives in Quantum Science, Education, and Industry Partnerships, 2026–present.** Contribute expertise in quantum hardware, photonics, sensing, and engineering education to FIU-wide research, workforce development, partnerships, and quantum-infrastructure planning.
- **FIU–IonQ quantum-computing initiative, 2026–present.** Participated in FIU’s faculty/leadership visit to IonQ and technical discussions on computing, networking, sensing, and education. Contribute to research and educational planning for FIU’s planned **IonQ Superion 256 trapped-ion quantum computer**. FIU announced the agreement in September 2026; campus arrival is expected in late 2027. FIU announcement.
- **Co-Founder, FIU Quantum Initiative, 2025–present,** with Prof. Janki Bhimani. Helped launch the Quantum Working Group and coordinate faculty capabilities, seminars, curriculum, and outreach. FIU CEC feature.
- **University–Industry Collaboration Session, FIU CEC Industry Program, November 20, 2025.** Presented “Quantum Photonics for Extreme Environments and Real-World Sensing” to academic and industry participants.

- **Statewide and industry engagement, 2025–present.** Invited Power Talk speaker at the Florida Quantum Summit (University of Florida, April 2025); organizer of FIU’s participation in Quantum Beach 2025; faculty representative for FIU Quantum AI at eMerge Americas 2026; participant in FIU–Microsoft quantum discussions (June 2026).

Professional Memberships

- **Institute of Electrical and Electronics Engineers (IEEE)** – Senior Member (since 2024); Member, IEEE Photonics Society.
- Member: Optica; American Physical Society (APS); SPIE; European Association on Antennas and Propagation (EurAAP); American Society for Engineering Education (ASEE).

Grants and Funding

Active Research Awards

1. **DOE Office of Science / Brookhaven National Laboratory: El-Pho.** *Electro-Photonics Integrated Platform for Near-Sensor Processing in Extreme Environments*; FIU subcontract **PI**; subcontract 463609. **\$600,000 ceiling**, incrementally funded (\$60,000 initially allotted). Base period: March–September 2026; annual options through September 30, 2029. **MEERCAT** project.
2. **AFOSR: Complex Frequency Excitations in Natural and Artificial Plasma.** **PI**; award FA9550-26-1-B070; **\$50,000**; June 1–November 30, 2026.
3. **NSF I-Corps: Meta-MRI.** *Translation Potential of Electromagnetic Wave Manipulation to Enhance Magnetic Resonance Imaging Systems*; **PI**; award 2433854; **\$50,000**; July 1, 2024–December 31, 2026.

Completed and Recent Research Support

- **ONR: Next Generation of Systems Enabled by Nanoscale Coatings.** Co-I / Senior Personnel; \$546,273 total award; 2023–2025.
- **DOE SBIR Phase I / RadiaBeam: Ferrite-Free Circulator for Precise Measurements of SRF Cavities with High Q-Factor.** Co-PI / FIU lead; \$63,000 to FIU; 2022–2023.
- **IC Postdoctoral Research Fellowship: Quantum Sensors for Optical Readout of Superconducting Qubits.** Research Advisor / Host for Dr. Ingrid Torres; October 2023–January 14, 2026.
- **FIU: Hands-On Photonics and Quantum Technologies Educational Kits.** **PI**; \$10,000; 2023–2024.

Institutional Start-up Support

- **FIU faculty start-up package, 2021:** **\$500,000** for research and infrastructure over the first two years, plus separate tuition and stipend support for **three graduate students for two academic years**.

Industry Computing Resources

- **NVIDIA Academic Grant Program: CFQ-Control GPU Compute Award.** PI; 32,000 A100 80GB GPU-hours via NVIDIA Brev; January–June 2026.
- **IonQ Quantum Computing Access.** Cloud quantum-computing access for teaching and research.

Teaching Experience

Florida International University

August 2021 – Present

Developed and taught EEE 4423 *Introduction to Quantum Computers* and EEE 6429 *Advanced Quantum Computers*. Other teaching includes EEL 6020 *Numerical Analysis of Electrical Devices*, special topics in quantum technologies and nanophotonics, dissertation research, and senior design.

Links: EEE 4423 (ECE news); EEE 4423 (FIU catalog); EEE 6429 (ECE news); EEE 6429 (flyer).

Courses use Qiskit, Python, Jupyter notebooks, full-wave simulation, recorded lectures, and free teaching materials. The QTM Lab YouTube channel contains 70+ recorded modules and complete course playlists.

Textbook. Author of *Introduction to Quantum Computing and Sensing for Electrical Engineers: A Hands-On Guide* (Springer, Synthesis Lectures on Emerging Engineering Technologies, forthcoming 2026; Springer page). Based on EEE 4423 and EEE 6429, the book includes Qiskit and Python exercises.

Student Evaluations (FIU SPOT, 2021–2026)

Category and overall averages are reported by FIU on a 5-point scale (Excellent = 5; Very Good = 4; Good = 3; Fair = 2; Poor = 1). Across 16 evaluated sections, **243 of 263 enrolled students responded (92.4% response rate)**.

Term	Course	Sec.	Resp./ Enr.	Struct.	Supp.	Inter.	Overall
Fall 2021	EEL 6020 Numerical Analysis of Electrical Devices	U01	29/34	2.87	2.83	2.83	2.84
Spring 2022	EEL 6931 Quantum Technologies (Special Topics)	U22	1/4	3.33	4.00	4.50	3.88
Fall 2022	EEL 6020 Numerical Analysis of Electrical Devices	U01	30/32	3.89	3.85	4.07	3.92
Spring 2023	EEE 6429 Advanced Quantum Computers	RVC	5/5	4.53	4.60	4.70	4.60
Fall 2023	EEE 4423 Introduction to Quantum Computers	RVC	18/21	3.91	3.96	4.06	3.97
Fall 2023	EEL 6020 Numerical Analysis of Electrical Devices	U01	16/17	4.48	4.40	4.56	4.47
Spring 2024	EEE 4423 Introduction to Quantum Computers	RVC	10/10	4.07	4.07	4.35	4.14
Spring 2024	EEE 6429 Advanced Quantum Computers	RVC	14/14	4.26	4.38	4.46	4.36
Fall 2024	EEE 4423 Introduction to Quantum Computers	RVC	7/7	4.67	4.71	4.71	4.70

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Student evaluations (continued)

Term	Course	Sec.	Resp./ Enr.	Struct.	Supp.	Inter.	Overall
Fall 2024	EEL 6020 Numerical Analysis of Electrical Devices	U01	22/22	4.71	4.55	4.82	4.68
Spring 2025	EEE 4423 Introduction to Quantum Computers	RVC	21/23	4.37	4.10	4.22	4.23
Spring 2025	EEE 6429 Advanced Quantum Computers	RVC	12/13	4.64	4.53	4.54	4.57
Fall 2025	EEE 4423 Introduction to Quantum Computers	RVC	4/5	4.58	4.58	4.63	4.59
Fall 2025	EEL 6020 Numerical Analysis of Electrical Devices	U01	37/39	4.57	4.50	4.64	4.56
Spring 2026	EEE 6429 Advanced Quantum Computers	RVC	12/12	4.22	4.06	4.17	4.15
Spring 2026	EEE 4423 Introduction to Quantum Computers	RVC	5/5	—	—	—	3.43 [†]

[†]Overall rating from Faculty180. Category ratings are unavailable separately for EEE 4423; the SPOT export combines both Spring 2026 quantum-computing courses.

EEL 6020 student evaluations. Overall ratings increased from **2.84 in Fall 2021 to 4.56 in Fall 2025**, reaching **4.68 in Fall 2024**; enrollment grew from 34 to 39. Course changes included recorded lectures, graded Ansys HFSS simulation labs in place of exams, and weekly simulation tutorials. The median overall rating across all 16 evaluated sections is 4.30, and 4.46 for sections taught since 2024.

Student feedback highlights recorded lectures, practical assignments, Python/Qiskit exercises, and Ansys HFSS labs. Requested additions include worked programming examples, applications, and short quizzes.

External Learning Assessment

Administered the Quantum Computing Conceptual Survey (QCCS) in EEE 4423 in Spring 2026, with comparisons to 878 students in 59 courses at 46 institutions. Survey results, student comments, and assignment performance inform course updates.

Selected Education Outreach and Public-Facing Teaching

- **FIU Science & Fiction Lab Interdisciplinary Faculty Fellowship Institute**, 2026 cohort.
Selected as a Faculty Fellow. Developing an assessed jigsaw module that links an engineering concept to a fiction-based analogy. Planned implementation: Fall 2026.
- **FIU Qiskit Fall Fest 2025**, Florida International University, Nov. 8–10, 2025.
Organizer and host of a three-day quantum-computing event with workshops, challenges, mentoring, speakers, and judging.
Links: [FIU calendar listing](#); [program site \(GitHub\)](#).
- **Florida Quantum Summit 2025**, University of Florida, Gainesville, FL, Apr. 17, 2025.
Invited “Power Talk” speaker on quantum computing, sensing, and workforce development.
Links: [event page](#); [agenda](#).
- **IBM Quantum Computing Fall Fest**, Miami Dade College, Nov. 2024.
Panelist and mentor for quantum-algorithm workshops and career discussions.

- **International Lecture Series on Nanophotonics**, Ludwig-Maximilians-Universität München, Oct. 2022.
Invited opening lecture: “Anomalies of Light Scattering in Low-Symmetry Nanophotonics.”
- **FIU KFSCIS Seminar**, Florida International University, Mar. 2022.
Invited seminar: “Scattering Anomalies: Recent Breakthroughs and Quantum Applications.”
Link: recording (YouTube).

Courses at FIU

EEE 4423 (RVC): Introduction to Quantum Computers

Links: ECE news; FIU catalog.

- Undergraduate course on quantum computing, quantum information, quantum circuits, Qiskit/Python workflows, and physical implementations.
- Recognized with the *Affordability Counts* medallion for zero-textbook-cost course design.
- Offered: Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026.

EEE 6429 (RVC): Advanced Quantum Computers

Links: ECE news; flyer.

- Graduate course on quantum-computing architectures, algorithms, physical implementations, and error-resilient control.
- Recognized with the *Affordability Counts* medallion.
- Offered: Spring 2023, Spring 2024, Spring 2025, Spring 2026.

EEL 6020 (U01): Numerical Analysis of Electrical Devices

Links: course description; FIU catalog; QTM Lab playlist.

- Graduate course on finite differences, finite elements, integral methods, field modeling, device simulation, and engineering interpretation.
- Offered: Fall 2021, Fall 2022, Fall 2023, Fall 2024, Fall 2025.

EEL 6931: Quantum Technologies and Materials / Special Topics

- Graduate special-topics course on quantum sensing, communication, computation, superconductors, 2D materials, metasurfaces, and complex-frequency wave physics.
- Offered: Spring 2022, Spring 2026.

Special Topics: Quantum Nanophotonics

- Specialized course on quantum mechanics and nanophotonics, including embedded eigenstates, bound states in the continuum, non-Hermitian scattering, and complex-frequency excitation.
- Offered: Summer 2024.

Senior Design I and II: Faculty Project Mentorship

- *Low-Cost Air Quality Kit*, Fall 2023–Spring 2024: five-student team developing a low-cost air-quality monitoring system.
- *Handheld Moisture Tomography Device for Wall Inspection*, 2026–present: four-student team developing radar-based, non-destructive moisture mapping behind drywall. Senior Design I, Spring 2026; Senior Design II, Fall 2026.

EEL 7980: Ph.D. Dissertation

- Dissertation supervision through arranged meetings, research milestones, manuscript development, and placement planning.
- Offered: Summer 2025 and Fall 2025.

Editorial Work and Conference Organization

Journal Editorial Roles

- **Associate Editor**, *Photonics Research*, Optica Publishing Group / Chinese Laser Press, 2022–present. Handle manuscripts in nanophotonics, metasurfaces, quantum photonics, and electromagnetic wave physics.
Link: Photonics Research editorial board.
- **Editorial Board Member**, *Advanced Photonics Nexus*, SPIE / Chinese Laser Press, 2024–present.
Links: SPIE editorial board; APN journal hub.
- **Editorial Board Member**, *Reviews on Advanced Materials Science*, De Gruyter Brill, 2023–present.
Link: RAMS journal page.

Guest Editorships and Special Collections

- **Guest Editor**, *npj Metamaterials*, collection “Programmable and Intelligent Metamaterials for Sensing, Wireless Power, and Communications.” Co-editors: Jensen Li, Natalia Litchinitser, and Zin Lin. Collection and guest editors.
- **Guest Editor**, joint theme issue “Emerging Trends in Photonic Quantum Computing,” *Advanced Photonics* and *Advanced Photonics Nexus*, 2025–2026.
Co-editor: Prof. Xiulai Xu.
Links: Advanced Photonics call; APN call; Researching.cn hub.
- **Lead Guest Editor**, virtual feature issue “Physics of Complex Photonic Media and Metamaterials,” *Optical Materials Express*, Optica, 2023.
Co-editors: Viktoriia Babicheva and Carsten Rockstuhl.
Link: OME virtual feature issue.
- **Guest Editor**, special issue “Optical Nanoantennas,” *Photonics*, MDPI.
Link: MDPI special issue.

Conference and Community Organization

- **Organizer and Session Co-Chair**, “Complex Frequency Excitation Effects in Photonics and Wave Physics,” IEEE RAPID 2025, Miramar Beach, FL, Aug. 13–15, 2025.
Co-organized the focus session with Seunghwi Kim, CUNY ASRC.
Link: IEEE RAPID.
- **Organizer and Host**, FIU Qiskit Fall Fest 2025, Florida International University, Miami, FL, Nov. 8–10, 2025.
Led a three-day quantum-computing event with workshops, challenges, mentoring, speakers, and judging.
Links: FIU event listing; program repository.

- **FIU Representative and Organizer**, Quantum Beach 2025: Igniting Florida's Role in the Quantum Revolution, West Palm Beach, FL, Oct. 8, 2025.
Coordinated FIU participation in a statewide quantum event for business, policy, education, and community stakeholders.
Link: [Quantum Beach 2025](#).
- **Session Co-Chair**, "Quantum Techniques for Electromagnetics," sessions B14-1 and B14-2, 3rd URSI Atlantic Radio Science Meeting / Asia-Pacific Radio Science Conference (AT-AP-RASC), Gran Canaria, Spain, June 2, 2022.
- **Session and Roundtable Co-Chair**, 12th RASA-America Conference, St. Petersburg, FL, Nov. 20–21, 2021.
Co-chaired the Career Development Roundtable, Early Career Researchers session, and Physics 2 session.

Media Coverage and Public Engagement

Authored Commentary in the Popular and Trade Press

- **A. Krasnok**, "Universities should judge quantum-computing investments by what students learn," *Physics World*, September 9, 2026. [Physics World article](#).
- **"Quantum sensors could spot hidden damage in the thousands of US bridges rated structurally deficient,"** *The Conversation*, June 19, 2026. Republished by *Scientific American* (June 25, 2026) as "How Quantum Sensing Could Reveal Hidden Faults in Thousands of U.S. Bridges," and by *FIU News* (June 19, 2026).
Links: [The Conversation](#); [Scientific American](#); [FIU News](#).
- **"Quantum sensors use atoms, electrons and light as ultra-steady rulers,"** *The Conversation*, May 20, 2026. Republished by *FIU News* (May 20, 2026).
Links: [The Conversation](#); [FIU News](#).
- **"A New Knob for Resonant Photonics: The Waveform,"** *Optics & Photonics News* **37**, feature article, April 2026.
Link: [OPN feature](#).
- **"Cryogenic photonics gives superconducting quantum hardware a path to scale,"** *Laser Focus World*, May 7, 2026.
Link: [Laser Focus World](#).
- **"Universities Need Transparent Quantum Hardware,"** *Nature Reviews Physics*, Comment, July 15, 2026 (with A. Sarwat, W. Anderson Jr., and A. Gil).
Link: [doi:10.1038/s42254-026-00963-4](https://doi.org/10.1038/s42254-026-00963-4).
- **"Russia: scientists petition to end political persecution,"** *Nature* **591**(7849), 202 (2021), Correspondence (with A. Kabanov and D. Seletskiy).
Link: [doi:10.1038/d41586-021-00594-y](https://doi.org/10.1038/d41586-021-00594-y).

Research and Recognition Covered by Third Parties

- **Physics World Top 10 Breakthroughs of 2020** — photonic twistronics and twisted α -MoO₃ polaritons.
Links: Physics World; CUNY ASRC news.
- **Optics & Photonics News, “Optics in 2018” annual feature selection** — coherent optics for energy transfer, storage, and release.
Link: OPN Optics in 2018.
- **Physics (APS) highlight** — accompanying “Coherently Enhanced Wireless Power Transfer,” *Phys. Rev. Lett.* **120**, 143901 (2018), an Editors’ Suggestion.
Link: doi:10.1103/Physics.11.33.
- **FIU CEC, “From the Dean’s Desk: The New Wave of Quantum Research,”** Nov. 2025.
Link: CEC feature.
- **FIU News, “As if you were there: See FIU at eMerge Americas 2026,”** Apr. 2026.
Link: FIU News recap.
- **FIU ECE Newsroom** — FIU Top Scholar 2025; Leopold B. Felsen Award 2024; launch of EEE 4423 and EEE 6429.
Links: Top Scholar; Felsen Award; EEE 4423; EEE 6429.
- **Florida Alliance for Quantum Technology (FAQT) meeting recap**, USF, Sept. 2025.
Link: FAQT recap.
- **FloridaQuantum.org Academic Hub** — FIU research profile listing.
Link: FloridaQuantum Academic Hub.

Open Educational Resources and Digital Scholarship

- **QTM Lab YouTube channel** — 70+ recorded lecture modules and complete course playlists covering quantum computing, quantum sensing, and computational electromagnetics, openly available to students and the public.
Links: channel; EEL 6020 playlist.
- **Zero-textbook-cost course design** — EEE 4423 and EEE 6429 both recognized with the Florida *Affordability Counts* medallion for eliminating required textbook costs.
- **Qiskit Fall Fest program materials** — open program repository of workshop notebooks, challenges, and organizing materials from FIU Qiskit Fall Fest 2025.
Link: program repository.
- **QTM Lab website and publication archive** — group research pages, open publication list, and student-facing resources.
Link: krasnokqtm.com.
- **Research preprints** — manuscripts available on arXiv; publication records maintained through ORCID.

Supervision and Mentorship

Mentoring focus: modeling, simulation, device design, measurement, validation, publication, reproducible code, and career placement.

Record: Three postdoctoral fellows supervised, one Ph.D. co-supervised to completion, several graduate researchers advised, and 11 undergraduate researchers and Senior Design students mentored at FIU. Deepanshu Trivedi graduated in Dec. 2025 and joined **imec USA Florida** as a Design Engineer on the Superconducting Electronics team; Dr. Muhammad Mahmudul Hasan moved to **Nokia** in Aug. 2026; earlier mentees placed at **NTU Singapore**, **DTU Electro**, the **University of Florida**, and industry engineering roles.

Postdoctoral Fellows

- **Ingrid Torres, Ph.D.** – Postdoctoral Fellow, FIU, 2024–present.
Focus: quantum sensing, nanofabrication, cryogenic dielectric resonators, quantum photonics, metasurfaces, and cryogenic wireless sensing. Selected outputs: “Cryogenic dielectric Mie resonators for quantum applications,” *Proc. SPIE* **13379**, 2025, doi:10.1117/12.3052180; “A Cryogenic Dielectric Antenna for Wireless Sensing and Interfacing Outside the 10 K Environment,” arXiv:2509.06199, doi:10.48550/arXiv.2509.06199.
- **Muhammad Mahmudul Hasan, Ph.D.** – Postdoctoral Research Associate, FIU, 2025–2026.
Focus: quantum sensors, THz photonics, diamond TeraFETs, mid-IR/THz devices, and measurement protocols. Placement: **Nokia** (departed August 1, 2026).
- **Asif Hassan, Ph.D.** – Postdoctoral Fellow, FIU, alumnus.
Focus: antenna design, microwave photonics, RF prototyping, measurements, and student training. Placement: Antenna Engineer in industry.

Graduate Alumni

- **Deepanshu Trivedi** – Ph.D., Electrical & Computer Engineering, FIU, Dec. 2025.
Co-supervised with Prof. Arjuna Madanayake. Current position: Design Engineer, imec USA Florida, Superconducting Electronics team. Focus: complex-frequency control, circuit/cavity QED, Fano-qubits, and scattering anomalies. Selected first-author outputs: “Fano-qubits for quantum devices with enhanced isolation and bandwidth,” *Appl. Phys. Lett.* **122**, 264003, 2023, doi:10.1063/5.0151047; “Anomalies in light scattering: A circuit-model approach,” *Phys. Rev. Applied* **22**, 034061, 2024, doi:10.1103/PhysRevApplied.22.034061; “Revealing invisible scattering poles with complex-frequency signals,” *J. Appl. Phys.* **137**, 243103, 2025, doi:10.1063/5.0262971. Professional link: [LinkedIn](#).
- **Md. Ibrahim Khalil** – Graduate Researcher, FIU, 2025–January 2026.
Focus: silicon microring resonators, electromagnetic solver cross-validation, temporal coupled-mode modeling, and free-carrier effects in silicon photonics. Transferred to another university in Spring 2026; completed the project handover in January 2026.

Current Graduate Researchers and Research Collaborators

- **Aditya Bhowmik** – Graduate Researcher, FIU, 2025–present.
Focus: complex-frequency wave physics, microwave and optical interfaces, system modeling, and measurements.

- **Md. Alim Uddin** – Graduate Research Assistant, FIU, May 2026–present.
Focus: quantum photonics, electromagnetic modeling, metamaterial-enabled sensing, and computational methods.
- **Md. Nakibul Islam** – Graduate Research Assistant, FIU, August 2026–present.
Focus: quantum photonics, electrodynamics, metamaterials, and computational modeling.
- **Laraib Niaz** – Research collaborator and Ph.D. research mentee, 2024–present.
FIU Ph.D. readmission pending; continuing remote research collaboration. Focus: quantum simulations of correlated phases and complex-frequency control in many-body systems. Co-author on “Quantum leap: observing antiferromagnetic transition in a 3D fermionic Hubbard model with ultracold atoms,” *Advanced Photonics* **6**, 060503, 2024.

Undergraduate Research and Senior Design Mentorship

Undergraduate Researchers

- **Nafiul Khalid** – Undergraduate researcher in Computer Science, FIU Honors College, 2025–present.
Focus: Qiskit and PennyLane programming, quantum machine learning, and development of quantum-computing textbook learning resources.
- **Quesnay Lundy** – Undergraduate research intern and mentee, 2023–2024.
Focus: wireless power transfer, transmitter–receiver coil measurements, and CST electromagnetic simulations. Continued studies at the University of Florida in 2024; Founder & CEO, EcoPulse Foundation.

Senior Design Projects

- **Handheld Moisture Tomography Device for Wall Inspection** – Faculty Mentor, FIU, 2026–present.
Students: Sean Avendano, Andy Lara, Pablo Lucero, and Degle Martinez.
Senior Design I, Spring 2026; Senior Design II, Fall 2026. Non-destructive moisture mapping behind drywall using a 60 GHz radar and guided two-dimensional scanning.
- **Low-Cost Air Quality Kit** – Faculty Mentor, FIU, Fall 2023–Spring 2024; completed.
Students: Caio Moreti Koga, Jasim Bhatti, Adrian Ortega, Andres Lara, and Nurain Chowdhury.
Senior Design I and II: development and testing of a low-cost air-quality monitoring system.

Selected Pre-FIU Mentorship

- **Sergey I. Lepeshov** – mentee, pre-2021.
Focus: hybrid nanophotonics and non-Hermitian metasurfaces. Selected joint outputs: “Switchable dual-mode nanolaser: mastering emission and invisibility through phase transition materials,” *Nanophotonics* **12**, 3729–3736, 2023; “Beyond bounds on light scattering with complex frequency excitations,” *Phys. Rev. Lett.* **129**, 203601, 2022. Placement: Postdoctoral Researcher, DTU Electro, Technical University of Denmark.
- **Guangwei Hu** – mentee, UT Austin era.
Focus: polaritonic nanophotonics and metasurfaces. Selected joint output: “Topological polaritons and photonic magic angles in twisted α -MoO₃ bilayers,” *Nature* **582**, 209–213, 2020. Placement: Nanyang Assistant Professor, Nanyang Technological University, Singapore.

Peer Review and Funding Review

- **Ad-hoc journal reviewer.** Review manuscripts in optics, photonics, applied electromagnetics, nanophotonics, materials, metamaterials, non-Hermitian systems, quantum technologies, and circuit/cavity-QED interfaces.
Record: ORCID: 0000-0001-7419-781X.
- **Funding reviewer.** Review proposals as invited for U.S. federal and international research programs, including the DOE Office of Science and National Science Foundation.
- **Conference reviewer and program service.** Review submissions and support technical programming for conferences in photonics, metamaterials, antennas, wave physics, and quantum technologies, including IEEE RAPID, Metamaterials Congress, and related meetings.

Institutional Service and Leadership

- **Co-Founder, FIU Quantum Initiative (FIU-QI),** 2025–present. Campus-wide coordination of quantum research, education, and workforce development with Prof. Janki Bhimani.
- **Founding participant, FIU Quantum Working Group,** 2025–present. Compiled an inventory of quantum-research faculty, facilities, and courses across FIU colleges.
- **Organizer and Host, FIU Qiskit Fall Fest 2025,** Nov. 8–10, 2025. Three-day campus quantum-computing event with workshops, challenges, mentoring, invited speakers, and judging.
- **Faculty Representative, FIU Quantum AI Showcase,** eMerge Americas, Miami Beach, Apr. 2026. Represented FIU quantum research to industry, investor, student, and public audiences; reviewed conference-floor communications materials.
- **FIU Representative, Quantum Beach 2025,** West Palm Beach, Oct. 8, 2025. Coordinated FIU participation in a statewide quantum event for business, policy, education, and community stakeholders.
- **Participant, FIU leadership and faculty site visit to IonQ,** College Park, MD, 2026. Contributed to technical and strategic discussions on collaboration in quantum computing, networking, sensing, education, and workforce development.
- **Presenter, Dean's Leadership Council,** College of Engineering & Computing, FIU, Feb. 19, 2026. Briefed college leadership on quantum-enabled infrastructure sensing and quantum-assisted optimization.
- **Faculty Fellow, FIU Science & Fiction Lab Interdisciplinary Faculty Fellowship Institute,** 2026 cohort. Developing an assessed interdisciplinary teaching module for Fall 2026 implementation.
- **Courtesy appointment, Knight Foundation School of Computing and Information Sciences,** FIU. Supports cross-college quantum computing research and student co-advising.
- **Director, Photonics Core Facility,** CUNY Advanced Science Research Center, 2018–2021. Oversaw instrumentation planning, operating procedures, training, scheduling, and laboratory safety for CUNY and external users.

Certifications and Professional Training

- **Research compliance and integrity.** Responsible Conduct of Research for Engineers (CITI, issued Oct. 2025); Social and Behavioral Responsible Conduct of Research (CITI, issued Jul. 2024); Export Controls (CITI, issued Jul. 2024); Human Subjects Research, Basic/Refresher Course (CITI, issued Dec. 2023); Conflicts of Interest (CITI, issued Aug. 2022); Hazard Communication (FIU EH&S, issued Jul. 2024).
Provider: CITI Program.
- **Grant development.** University of South Florida Summer Grant Writing Workshops Program, USF Research Development Institute, issued Aug. 2024.
Link: [USF Summer Grant Writing Workshops](#).
- **Quantum computing courses.** *Quantum Computing for Everyone – An Introduction* (Coursera, issued Sep. 2025); *Mathematical Foundations and Quantum Mechanics Essentials* (Coursera, issued Sep. 2025).

Technical Skills

- **Electrodynamics & wave physics.** Complex-frequency and non-Hermitian scattering; pole-zero analysis; temporal coupled-mode theory; circuit models; Green-function and modal methods; Mie scattering; BICs; exceptional points; Brewster and embedded eigenstates; inverse design of metasurfaces and nano-optical structures.
- **Quantum photonics & control.** Open-system modeling; Lindblad master equations; circuit/cavity QED; Fano-type interference; robust pulse shaping; crosstalk suppression; quantum-control workflows in Python, [QuTiP](#), and [Qiskit](#).
- **Computational modeling.** FEM, FDTD, MoM, multiphysics coupling, parametric sweeps, uncertainty analysis, SNR analysis, and surrogate modeling. Tools: CST Studio Suite, COMSOL Multiphysics, Ansys HFSS, Keysight ADS, Zemax OpticStudio, MATLAB, Mathematica, Python, NumPy, SciPy, and Matplotlib.
- **Experimental photonics & microwaves.** Free-space and fiber alignment; PL/Raman spectroscopy; confocal mapping; microwave resonator measurements; VNA/S-parameter characterization; cryogenic and RF workflows; Python-based instrument control.
- **RF, resonators & antennas.** Dielectric and superconducting resonators; antenna matching; packaging and impedance control; near-/far-field characterization; model-measurement correlation; waveform-based diagnostics.
- **Reproducible research.** Git, $\text{\LaTeX}$, Python notebooks, documented data pipelines, provenance tracking, reusable scripts, and publication-quality visualization.

Leadership and Professional Skills

- **Laboratory and program leadership:** founding director of the CUNY ASRC Photonics Core Facility and director of FIU's QTM Lab, with responsibility for instrumentation, shared access, user training, and student mentoring.
- **Team culture and career development:** build respectful, supportive research teams through technical training, open feedback, and individual career guidance. Support researchers' progress within the group and their moves to new academic and industry positions.
- **Research and partnerships:** lead sponsored projects and coordinate theory, computation, experiments, and deliverables with university, national-laboratory, and industry partners.
- **Education and community:** develop quantum-engineering curricula and open teaching resources; organize workshops and outreach; serve in journal editorial roles and university-wide quantum planning.

Scientific Publications

Publications in wave physics, nanophotonics, quantum technologies, and metamaterials. Google Scholar (September 25, 2026): **15,222 citations**, *h-index* **61**, *i₁₀-index* **113**. Google Scholar; ORCID.

137 journal articles; 42 conference proceedings; 9 books and chapters, including one forthcoming book; 12 preprints and submitted manuscripts; 6 science-communication articles.

Science Communication and Commentary

1. **A. Krasnok**, "Universities should judge quantum-computing investments by what students learn," *Physics World*, September 9, 2026. *Physics World* article.
2. **A. Krasnok**, "Quantum sensors could spot hidden damage in the thousands of US bridges rated structurally deficient," *The Conversation* (June 19, 2026). *The Conversation* article.
Republished by *FIU News* (June 19, 2026), *FIU News* republication.
Republished as "How Quantum Sensing Could Reveal Hidden Faults in Thousands of U.S. Bridges," *Scientific American* (June 25, 2026), *Scientific American* republication.
3. **A. Krasnok**, "Quantum sensors use atoms, electrons and light as ultra-steady rulers, detecting faint motion, magnetism and gravity for navigation, medicine and science," *The Conversation* (May 20, 2026). *The Conversation* article.
Republished by *FIU News* (May 20, 2026), *FIU News* republication.
4. **A. Krasnok**, "Cryogenic photonics gives superconducting quantum hardware a path to scale," *Laser Focus World* (May 7, 2026).
URL: laserfocusworld.com/quantum/article/55375505.
5. **A. Krasnok**, "A New Knob for Resonant Photonics: The Waveform," *Optics & Photonics News* **37** (April 2026).
URL: optica-opn.org/home/articles/volume_37/april_2026.

6. A. Kabanov, **A. Krasnok**, and D. Seletskiy, "Russia: scientists petition to end political persecution," *Nature* **591**(7849), 202 (2021) (*Correspondence*).
DOI: 10.1038/d41586-021-00594-y.

Preprints and Submitted Manuscripts

1. **A. Krasnok**, "Which pulse maximizes resonant nonlinear conversion?" arXiv:2608.19464 (2026). Submitted to *Physical Review Applied*, September 2026.
2. **A. Krasnok**, "What is a quantum complex scattering zero?" Submitted to *Quantum Science and Technology*, August 16, 2026.
3. **A. Krasnok**, "Steering Light Through the Complex-Frequency Plane." Submitted feature, *Optics & Photonics News, Optics in 2026*, August 24, 2026.
4. **A. Krasnok**, "Quantum Computing Courses Need Real Machines." *Communications of the ACM*, under consideration.
5. S. Asgari, **A. Krasnok**, A. Lavrinenko, M. Khalily, H. Lipsanen, W. Whittow, P. Kuzhir, V. Asadchy, M. J. Huttunen, P. J. Soh, F. Ferranti, and T. Fabritius, "Graphene-Based Metamaterials: Fundamentals, Modeling Approaches, Emerging Trends, and Challenges." Submitted to *Laser & Photonics Reviews*, May 26, 2026.
6. **A. Krasnok**, "Protecting Qubits from Purcell Decay via Permanent Dipoles," *arXiv preprint* arXiv:2606.22732 (2026), doi:10.48550/arXiv.2606.22732.
7. I. Torres, R. Rodriguez, R. W. Laird, A. R. Laird, and **A. Krasnok**, "Twist-Tuned Bilayer Metasurface for 3 T MRI," *Journal of Physics D: Applied Physics*, under consideration. arXiv:2603.22639 (2026).
8. **A. Krasnok**, "From False Roots to Phasors: Negative and Complex Numbers in Mathematics, Physics, and Electrical Engineering," *arXiv preprint* arXiv:2603.21445 (2026), doi:10.48550/arXiv.2603.21445.
9. **A. Krasnok**, "Robust Superdirectivity in Electrically Small and Compact Antennas," *IEEE Antennas and Propagation Magazine*, under consideration. arXiv:2602.04121 (2026).
10. **A. Krasnok**, "Purcell-Like Environmental Enhancement of Classical Antennas: Self and Transfer Effects," *arXiv preprint* arXiv:2512.22363 (2025), doi:10.48550/arXiv.2512.22363.
11. O. Hemmatyar, S. Abdollahramezani, I. Zeimpekis, S. Lepeshov, **A. Krasnok**, A. I. Khan, K. M. Neilson, C. Teichrib, T. Brown, E. Pop, D. W. Hewak, M. Wuttig, A. Alù, O. L. Muskens, and A. Adibi, "Enhanced Meta-Displays Using Advanced Phase-Change Materials," arXiv:2107.12159 (2021).
12. D. G. Baranov, M. I. Petrov, and **A. E. Krasnok**, "Decoupling light and matter: permanent dipole moment induced collapse of Rabi oscillations," arXiv:1611.06897 (2016).

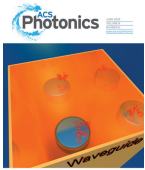
Published Journal Articles

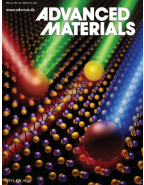
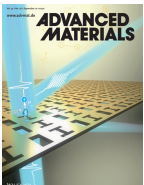
1. D. Trivedi, A. Bhowmik, L. Niaz, and **A. Krasnok**, "Reflection-Zero Waveforms for Selective Loading in Shared-Line Superconducting Qubit Circuits," *Advanced Quantum Technologies* **9**(9), e70441 (2026). doi:10.1002/qute.70441. Earlier version: arXiv:2506.03316. *Selected for a cover feature.*

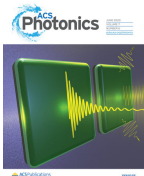
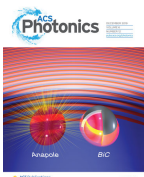
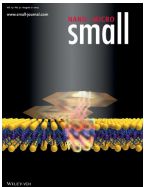
2. M. M. Hasan, I. Torres, and **A. Krasnok**, “Quantum Magnetometers for Infrastructure Inspection and Monitoring,” *Measurement* **290**, 122735 (2026), doi:10.1016/j.measurement.2026.122735.
3. **A. Krasnok**, “What Quantum Computer to Buy?” *AVS Quantum Science* **8**(3), 034101 (2026). doi:10.1116/5.0337237.
Preprint: arXiv:2604.04761.
4. **A. Krasnok** and D. Seletskiy, “Complex-Frequency Chirped Pulses for Trajectory-Resolved Scattering,” *ACS Photonics* (2026), doi:10.1021/acsphotonics.6c00602.
5. **A. Krasnok**, “Constant-Amplitude 2π Phase Modulation from Topological Pole–Zero Winding,” *Physical Review Letters* **137**(5), 053801 (2026), doi:10.1103/xt46-rfjs.
6. **A. Krasnok**, A. Sarwat, W. Anderson Jr., and A. Gil, “Universities Need Transparent Quantum Hardware,” *Nature Reviews Physics* (2026) [Comment], published July 15, 2026, doi:10.1038/s42254-026-00963-4.
7. I. Torres and **A. Krasnok**, “From Flat-Optics Concept to Qualified Hardware: Skills Map for the Meta-Optics and Diffractive Optics Workforce,” *Optical Engineering* **65**(10), 102007 (2026), doi:10.1117/1.OE.65.10.102007.
8. **A. Krasnok**, “Metamaterials in Superconducting and Cryogenic Quantum Technologies,” *Applied Physics Reviews* **13**, 021311 (2026), Special Collection: *Materials for Superconducting Quantum Devices*, doi:10.1063/5.0282013.
9. **A. Krasnok** and X. Xu, “Editorial: Emerging Trends in Photonic Quantum Computing,” *Advanced Photonics* **8**(1), 010102 (2026), doi:10.1117/1.AP.8.1.010102.
10. **A. Krasnok**, “Coherently assisted wireless power transfer through poorly transparent barriers,” *Journal of Physics D: Applied Physics* **59**(12), 125501 (2026), doi:10.1088/1361-6463/ae51a1.
11. **A. Krasnok**, “Physics paradoxes as stress tests from tea leaves to bell tests,” *The European Physical Journal Plus* **141**(3), 273 (2026), doi:10.1140/epjp/s13360-026-07519-6.
12. I. Torres and **A. Krasnok**, “Twist-Tuned Strong Coupling in Sub-GHz Wire Metasurface Bilayers,” *Microwave and Optical Technology Letters* **68**(3), e70540 (2026), doi:10.1002/mop.70540.
13. I. Torres and **A. Krasnok**, “A cryogenic dielectric antenna for wireless sensing and interfacing outside the 10 K environment,” *Cryogenics* **155**, 104286 (2026), doi:10.1016/j.cryogenics.2026.104286.
14. D. Trivedi, L. Niaz, A. Alù, and **A. Krasnok**, “Selective Excitation of Coupled Resonators via Complex Frequency Driving: Enhanced Efficiency and Crosstalk Suppression,” *ACS Photonics* **12** (11), 5929–5938 (2025), doi:10.1021/acsphotonics.5c01244.
15. A. C. Araujo-Martinez, **A. Krasnok**, S. V. Kutsaev, A. H. Seltzman, and A. Yu. Smirnov, “Virtual Critical Coupling in High-Power Resonant Systems,” *IEEE Transactions on Plasma Science* **53**(9), 2410–2418 (2025), doi:10.1109/TPS.2025.3590305.
16. D. Trivedi, A. Madanayake, and **A. Krasnok**, “Revealing Invisible Scattering Poles with Complex-Frequency Signals,” *Journal of Applied Physics* **137** (24), 243103 (2025), doi:10.1063/5.0262971.
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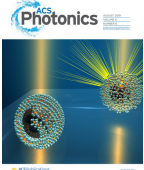
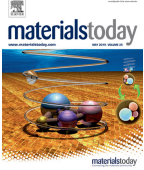
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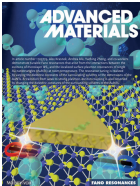
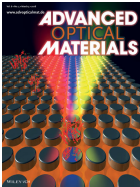


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Books and Book Chapters

1. **A. Krasnok**, *Introduction to Quantum Computing and Sensing for Electrical Engineers: A Hands-On Guide*. Synthesis Lectures on Emerging Engineering Technologies, Springer, Cham, first edition, forthcoming 2026. Hardcover ISBN: 978-3-032-34529-5; eBook ISBN: 978-3-032-34530-1. [Springer book page](#).
2. L. A. Jauregui, A. Ambrosio, L. Sortino, **A. Krasnok**, and F. M. Mayor, "Suppressing Material Loss in the Visible and Near-Infrared Range for Functional Nanophotonics Using Bandgap Engineering," in *Advances in Nanophotonics*, Chapter 3 (Scientific Research Publishing (SciRP), 2023). Book page: scirp.org/book/2968.
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Intellectual Property

U.S. Patent Applications

- **A. Krasnok**, D. Trivedi, and L. Niaz, “Systems and Methods for Selective Control of Coupled Quantum Elements,” U.S. Patent Application Serial No. 19/694,219, filed June 1, 2026 (FIU disclosure 2025-043; Wellspring record D2025-0055; docket FIU.632).
- **A. Krasnok** and I. Torres, “Tunable Metamaterial Architectures for Medical Imaging Applications,” U.S. Patent Application Serial No. 19/692,205, filed May 29, 2026 (FIU disclosure 2024-022; Wellspring record D2024-0023; docket FIU.627).
- **A. Krasnok** and A. Hassan, “Reconfigurable Antenna,” U.S. Patent Application Serial No. 19/692,154, filed May 29, 2026 (FIU disclosure 2025-008; Wellspring record D2025-0008; docket FIU.618).

FIU Invention Disclosures

- **A. Krasnok**, “Trajectory-Resolved Complex-Frequency Scattering Measurement with Chirped Analytic Pulses,” FIU invention disclosure D2026-0075 (2026; under review).
- **A. Krasnok**, “Topological Pole-Zero Winding for Constant-Amplitude Full-Cycle Phase Modulation,” FIU invention disclosure D2026-0074 (2026; under review).
- **A. Krasnok**, “Dressed-Rate-Optimized Permanent-Dipole Purcell Protection with Longitudinal Qubit Readout,” FIU invention disclosure D2026-0073 (2026; under review).
- **A. Krasnok**, “A Path Integral Framework for Systems with Memory and Foresight,” FIU invention disclosure D2025-0052 (2025; disclosure accepted for review).
- **A. Krasnok**, “Extreme Metasurfaces Enable Targeted and Protected Wireless Energy Transfer,” FIU invention disclosure D2022-0004 (2022; disclosure accepted for review).
- **A. Krasnok**, “Electromagnetic Power Tunneling via Epsilon-Near-Zero Metasurface,” FIU invention disclosure D2021-0084 (2021; disclosure accepted for review).

Invited Talks, Seminars, and Conference Presentations

Invited Talks, Seminars, and Colloquia

- **September 24, 2026** — Invited PAMS Seminar, Department of Physics, Astronomy and Materials Science, Missouri State University (virtual): “Quantum Computing through the Lens of Physics.”
- **Mar. 19, 2026** – Invited Guest Speaker, NewFoS Community College Workshop Series, virtual.
Talk: “Complex-Frequency Wave Phenomena: From Light and Microwaves to Acoustics and Elastic Waves.”
Link: NewFoS Community College Workshop Series.
- **Feb. 20, 2026** – Invited Presentation, imec USA Florida, NeoCity, Kissimmee, FL, USA.
Talk: “Cryogenic Quantum Systems Engineering: From mK Qubits to Practical RF/Photonics Control.”
Links: imec USA Florida; imec USA.
- **Feb. 19, 2026** – Presentation, Dean’s Leadership Council, College of Engineering & Computing, Florida International University, Miami, FL, USA.
Talk: “Quantum-Enabled Resilience: Infrastructure Health Sensing and Quantum-Assisted Optimization from Models to Prototypes and Testing.”
- **Dec. 2, 2025** – Invited Seminar, University of North Texas, Denton, TX, USA.
Talk: “Engineering Complex-Frequency Excitations: From Virtual Gain in Photonics to Selective Qubit Control.”
- **Aug. 18, 2025** – Invited Speaker, Florida Alliance for Quantum Technology / Florida Quantum Initiative Meeting, University of South Florida, Tampa, FL, USA.
Talk: “Quantum Sensing for Infrastructure Resilience: Testing at FIU’s Wall of Wind.”
Links: FAQT meeting recap; Florida quantum events listing.
- **May 13, 2025** – Seminar Speaker, Australian Research Council Centre of Excellence for Transformative Meta-Optical Systems (TMOS), hybrid seminar.
Talk: “Complex Frequency Excitations in Classical and Quantum Photonics.”
- **Apr. 17, 2025** – Invited Speaker, Florida Quantum Summit, University of Florida, Gainesville, FL, USA.
Power Talk: “From Quantum Light to Student Might: How FIU is Teaching and Building the Future of Sensing Technology.”
Links: event page; agenda.
- **Jan. 16, 2025** – Seminar Speaker, University of North Carolina at Charlotte.
Talk: “Complex Frequency Excitations in Classical and Quantum Photonics.”
- **Dec. 4, 2024** – Panelist, ASRC Alumni Community Virtual Roundtable, CUNY Advanced Science Research Center, virtual.
- **Nov 8–9, 2024** – Panelist, IBM Quantum Computing Fall Fest, Miami Dade College, Miami, FL, USA.
Panels and workshops on quantum algorithms, Qiskit, AI applications, and workforce development.
- **Oct 16, 2024** – CREOL Seminar, University of Central Florida, Orlando, FL, USA.
Talk: “Complex Frequency Excitations in Classical and Quantum Photonics.”
- **Oct 9, 2024** – Air Force Research Laboratory Seminar, USA.
Talk: “Complex Frequency Excitations in Photonics.”
- **Oct 1, 2024** – IIT Delhi Seminar, New Delhi, India.
Talk: “Complex Frequency Excitations in Photonics.”

- **Sept 12, 2024** – IO Seminar Series, Brookhaven National Laboratory (BNL), Upton, NY, USA.
Host: Soumyajit Mandal.
Talk: *“Unveiling the Invisible: Complex-Frequency Excitations in Photonics and Wave Physics.”*
- **June 24, 2024** – Air Force Research Laboratory Seminar, USA.
Talk: *“Complex Frequency Excitations in Photonics.”*
- **Dec 12, 2022** – Physics Seminar, Polytechnique Montréal, Montréal, QC, Canada.
Invited Lecture: *“Quantum Nonreciprocity with Nonlinearity and Weyl Semimetals.”*
Recording: YouTube.
- **Dec 2, 2022** – Institute of Photonics Technology, Jinan University, China (online seminar).
Invited Lecture: *“Anomalies in Light Scattering.”*
- **Oct 26, 2022** – International Lecture Series on Nanophotonics, Ludwig-Maximilians-Universität München (LMU), Munich, Germany.
Invited Opening Lecture: *“Anomalies of Light Scattering in Low-Symmetry Nanophotonics.”*
- **Aug 23–24, 2022** – SPIE Optics + Photonics, USA.
Invited Presentation: *“Low-Symmetry Nanophotonics.”*
- **May 20, 2022** – Invited Speaker, IEEE Photonics Society / IEEE Sensors Council Miami Section Joint Chapter, International Day of Light.
Talk: *“Quantum Embedded Superstates for Sensing Applications.”*
- **Mar 18, 2022** – FIU KFSCIS Seminar, Florida International University, Miami, FL, USA.
Talk: *“Scattering Anomalies: Recent Breakthroughs and Quantum Applications.”*
Recording: YouTube.
- **Feb 18, 2022** – Special Seminar, University of South Florida (USF), USA.
Talk: *“Broken-Symmetry Nanophotonics.”*
- **Feb 2, 2022** – Colloquium, University of Miami, Miami, FL, USA.
Talk: *“Scattering Anomalies: Recent Breakthroughs and Quantum Applications.”*
- **Jan 5, 2022** – USNC-URSI National Radio Science Meeting (NRSN), USA.
Invited Presentation: *“B3.6: Anomalies in Light Scattering for Advanced Wireless Power Transfer.”*

Contributed Conference Presentations (Selected)

- **Nov. 16, 2024** – XV RASA-America International Conference, University of North Carolina at Chapel Hill, Chapel Hill, NC, USA.
Talk: *“Complex Frequency Excitations in Classical and Quantum Photonics.”*
- **Oct. 18, 2024** – Florida Defense Science and Technology Symposium, University of Central Florida, Orlando, FL, USA.
Poster: *“Complex-Frequency Excitations and Quantum Sensors for Defense Applications.”*
- **Aug 15, 2024** – 2024 IEEE Research and Applications of Photonics in Defense Conference (RAPID), Miramar Beach, FL, USA.
Talk: *“Circuit Theory for Anomalies in Light Scattering.”*
- **June 2, 2022** – 3rd URSI Atlantic Radio Science Meeting / Asia-Pacific Radio Science Conference (AT-AP-RASC), Gran Canaria, Spain.
Talk: *“Quantum Nonreciprocity with Nonlinearity and Weyl Semimetals.”*

- **Nov. 21, 2021** – 12th RASA-America Conference, St. Petersburg, FL, USA.
Talk: *“Anomalies in Light Scattering for Modern Photonic Devices.”*
- **Feb 20–21, 2020** – Graphene & 2D Materials International Conference and Exhibition, New York, NY, USA.
Talk: *“Hybrid Metasurfaces for Enhanced Light–Matter Interactions and Extreme Polariton Manipulation.”*
- **Sept 16–21, 2019** – 13th International Congress on Artificial Materials for Novel Wave Phenomena (Metamaterials 2019), Rome, Italy.
Contributed talks: *“PT-Symmetric Cladding Layers for High-Q Brewster Modes and Embedded Eigenstates.”*; *“Coherent Control of Scattering in Non-Hermitian PT-Symmetrical Systems.”*
- **Sept 15–19, 2019** – Frontiers in Optics 2019, Washington, DC, USA.
Talk: *“Coherently Driven Embedded Eigenstates.”*
- **May 5–10, 2019** – CLEO 2019, San Jose, CA, USA.
Talk: *“Coherent Virtual Absorption and Embedded Eigenstates in Non-Hermitian PT-Symmetrical Systems.”*
- **June 4–8, 2018** – International Conference Laser Optics (ICLO 2018), St. Petersburg, Russia.
Talks: *“Strong Coupling in Core-Shell Nanostructure Based on Silicon Nanoparticle and TMDC Monolayer.”*; *“All-Dielectric Metasurface for Enhanced Optical-to-Terahertz Conversion Efficiency in Photoconductive Antenna.”*
- **May 13–18, 2018** – CLEO: Science and Innovations 2018, San Jose, CA, USA.
Talk: *“Separating Valley Excitons in a MoS₂ Monolayer at Room Temperature with a Metasurface.”*
- **Mar 14, 2018** – SPIE OPTO 2018, San Francisco, CA, USA.
Talk: *“Unusual Scattering Features of Bound States in the Continuum in Open Structures.”*
- **Sept 21, 2017** – SPIE Nanoscience + Engineering 2017, San Diego, CA, USA.
Talk: *“Embedded Eigenstates and Virtual Absorption Using Metamaterials.”*
- **June 19–23, 2017** – Days on Diffraction (DD 2017), St. Petersburg, Russia.
Talk: *“Experimental Demonstration of a Reconfigurable Magnetic Fano Resonance in Hybrid Oligomers.”*
- **May 22–25, 2017** – Progress In Electromagnetics Research Symposium (PIERS 2017), St. Petersburg, Russia.
Contributed talks: *“Ultrafast Tunable Hybrid Yagi–Uda Nanoantenna”*; *“Plasmonic Nanoantenna for Enhancement of Vertical Emission from Whispering Gallery Mode Laser”*; *“Approach for Fine-Tuning of Hybrid Dimer Nanoantennas via Laser Melting.”*
- **May 14–19, 2017** – CLEO: QELS Fundamental Science 2017, San Jose, CA, USA.
Talk: *“Virtual Electromagnetic Absorption and Energy Storage by a Hermitian System via Complex Frequency Excitation.”*
- **May 14–19, 2017** – CLEO: Science and Innovations 2017, San Jose, CA, USA.
Talk: *“Nonlinear Core-Shell Yagi–Uda Nanoantenna for Highly Tunable Directive Emission.”*
- **Sept 19–22, 2016** – 10th International Congress on Advanced Electromagnetic Materials in Microwaves and Optics (Metamaterials 2016), Chania, Greece.
Contributed talks: *“Reconfigurable Metal–Dielectric Nanodimers as a Component of Hybrid Nanophotonics”*; *“Nonlinear All-Dielectric Nanoantenna Reconfigured by Electron–Hole Plasma.”*

- **June 26–July 1, 2016** – IEEE International Symposium on Antennas and Propagation (APSURSI 2016), Fajardo, Puerto Rico, USA.
Talk: *“Ultrafast Magnetic Light.”*
- **June 27–July 1, 2016** – Days on Diffraction (DD 2016), St. Petersburg, Russia.
Contributed talks: *“Raman Scattering Governed by Dark Resonant Modes in Silicon Nanoparticles”*; *“Tuning of Hybrid Nanostructures via Femtosecond Laser Reshaping at the Nanoscale”*; *“Reversible and Non-Reversible Tuning of Hybrid Optical Nanoresonators”*; *“Chiral Near-Field Formation with All-Dielectric Nanoantennas”*; *“Polarization and Angle-Dependent Enhancement of Raman Scattering from Silicon Nanodisks.”*
- **May 25–29, 2015** – Days on Diffraction (DD 2015), St. Petersburg, Russia.
Contributed talks: *“Fano Resonance in Chains of Dielectric Nanoparticles with Side-Coupled Resonator”*; *“Laser Writing of Nanoparticle-Based Plasmonic Structures”*; *“Input Impedance of Small Antenna Provides Purcell Factor.”*
- **Oct 6–9, 2014** – 44th European Microwave Conference, Rome, Italy.
Talk: *“All-Dielectric Optical ‘Huygens Source’.”*
- **May 26–30, 2014** – Days on Diffraction 2014, St. Petersburg, Russia.
Talk: *“Experimental Investigation of Magnetic Purcell Factor in Wire Metamaterials.”*
- **Aug 4–7, 2013** – SBMO/IEEE MTT-S International Microwave & Optoelectronics Conference (IMOC 2013), Rio de Janeiro, Brazil.
Talk: *“Superdirective Magnetic Nanoantennas with the Effect of Light Steering: Theory and Experiment.”*
- **Oct 14–16, 2013** – ICECom 2013, Dubrovnik, Croatia.
Talk: *“Superdirective Nanoantennas: Theory and Experiment.”*
- **July 7–13, 2013** – IEEE Antennas and Propagation Society International Symposium (APSURSI 2013), Orlando, FL, USA.
Talk: *“Ultra-Compact All-Dielectric Superdirective Antennas: Theory and Experiment.”*